

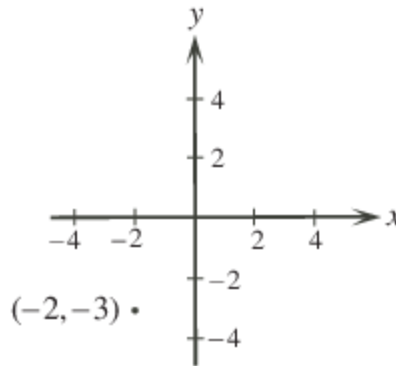


The CENTRE for EDUCATION
in MATHEMATICS and COMPUTING

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Topic Generator - Solution Set
Solutions

1. The point $(-2, -3)$ is reflected in the x -axis. What are the coordinates of its image after the reflection?



- (A) $(2, -3)$ (B) $(3, -2)$ (C) $(2, 3)$ (D) $(-3, -2)$ (E) $(-2, 3)$

Source: 2013 Gauss Grade 8 #9

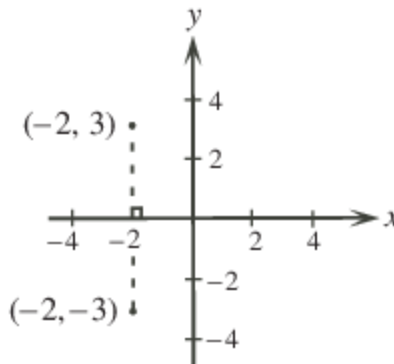
Primary Topics: Geometry and Measurement

Secondary Topics: Coordinate Geometry | Transformations

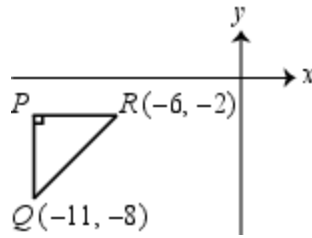
Answer: E

Solution:

After reflecting the point $(-2, -3)$ in the x -axis, the x -coordinate of the image will be the same as the x -coordinate of the original point, $x = -2$. The original point is a distance of 3 below the x -axis. The image will be the same distance from the x -axis, but above the x -axis. Thus, the image has y -coordinate 3. The coordinates of the image point are $(-2, 3)$.



2. In $\triangle PQR$ shown, side PR is horizontal and side PQ is vertical. The coordinates of P are



- (A) $(-8, -2)$ (B) $(-6, -8)$ (C) $(-11, -6)$ (D) $(-11, -2)$ (E) $(-8, -6)$

Source: 2016 Gauss Grade 8 #6

Primary Topics: Geometry and Measurement

Secondary Topics: Triangles | Coordinate Geometry

Answer: D

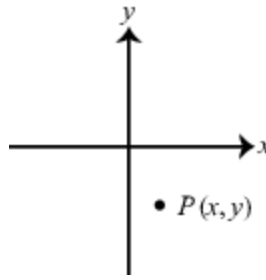
Solution:

Side PR is horizontal and so the y -coordinate of P is equal to the y -coordinate of R , or -2 .

Side PQ is vertical and so the x -coordinate of P is equal to the x -coordinate of Q , or -11 .

Therefore, the coordinates of P are $(-11, -2)$.

3. In the graph shown, which of the following statements is true about the coordinates of the point $P(x, y)$?



- (A) The values of both x and y are positive.
- (B) The value of x is positive and the value of y is negative.
- (C) The value of x is negative and the value of y is positive.
- (D) The values of both x and y are negative.
- (E) The value of x is 0 and the value of y is negative.

Source: 2021 Gauss Grade 8 #4

Primary Topics: Data Analysis

Secondary Topics: Graphs | Coordinate Geometry

Answer: B

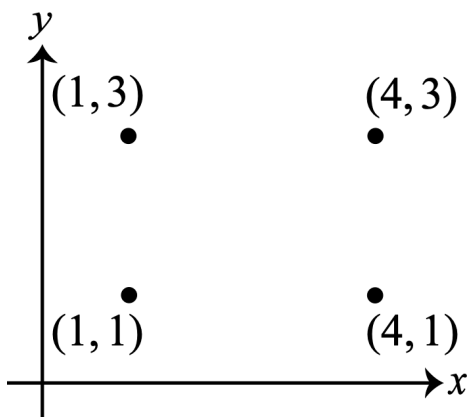
Solution:

Points which lie to the right of the y -axis have x -coordinates which are positive.

Points which lie below the x -axis have y -coordinates which are negative.

Point $P(x, y)$ lies to the right of the y -axis and below the x -axis and thus the value of x is positive and the value of y is negative.

4. The vertices of a rectangle have coordinates $(1, 3)$, $(1, 1)$, $(4, 1)$, and $(4, 3)$, as shown.



If the rectangle is reflected in the y -axis, which of the following points is *not* a vertex of the reflected rectangle?

- (A) $(-1, 1)$ (B) $(-4, 1)$ (C) $(-3, 4)$ (D) $(-1, 3)$ (E) $(-4, 3)$

Source: 2023 Gauss Grade 8 #10

Primary Topics: Data Analysis

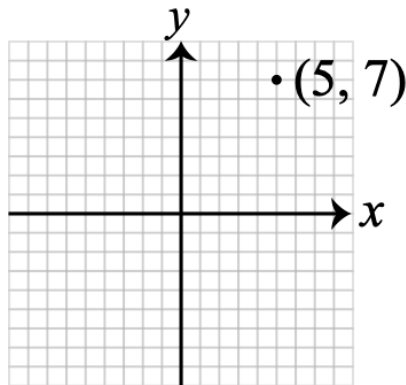
Secondary Topics: Graphs | Coordinate Geometry | Transformations

Answer: C

Solution:

When $(1, 3)$ is reflected in the y -axis, the reflected point is $(-1, 3)$. In general, when a point is reflected in the y -axis, its x -coordinate changes sign, and its y -coordinate does not change. Thus, the vertices of the reflected rectangle are $(-1, 3)$, $(-1, 1)$, $(-4, 1)$, and $(-4, 3)$. Of the given possibilities, $(-3, 4)$ is not a vertex of the reflected rectangle.

5. The point $(5, 7)$ is plotted on the graph shown.



When $(5, 7)$ is reflected in the x -axis, the resulting point is

- (A) $(5, -7)$ (B) $(-5, -7)$ (C) $(5, 7)$ (D) $(7, 5)$ (E) $(-7, -5)$

Source: 2025 Gauss Grade 8 #7

Primary Topics: Geometry and Measurement

Secondary Topics: Transformations | Coordinate Geometry

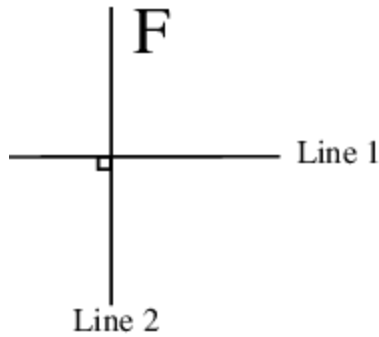
Answer: A

Solution:

When the point $(5, 7)$ is reflected in the x -axis, the resulting point is $(5, -7)$.

When a point is reflected in the x -axis, its x -coordinate does not change and its y -coordinate changes sign (provided that the original point is not on the x -axis).

6. The letter F is reflected in Line 1. The image is then reflected in Line 2. The shape that results is



- (A)  (B)  (C)  (D)  (E) 

Source: 2006 Gauss Grade 8 #15

Primary Topics: Geometry and Measurement

Secondary Topics: Transformations

Answer: D

Solution:

When the original shape



is reflected in Line 1 (that is, reflected vertically), the shape

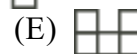
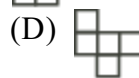
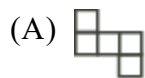


is obtained.

When this new shape is reflected in Line 2 (that is, reflected horizontally), the shape that results is



7. A 4×4 square grid can be entirely covered by three non-overlapping pieces made from 1×1 squares. If the first two pieces are  and  the third piece is



Source: 2008 Gauss Grade 8 #16

Primary Topics: Geometry and Measurement

Secondary Topics: Area | Transformations

Answer: C

Solution:

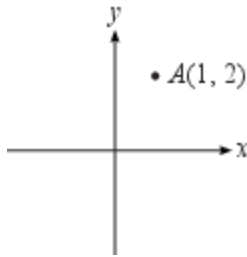
After some experimentation, the only way in which the two given pieces can be put together to stay within a 4×4 grid and so that one of the given choices can fit together with them is to rotate the second piece by 90° clockwise, and combine to obtain



Therefore, the missing piece is



8. The point $A(1, 2)$ is reflected in the y -axis. The new coordinates are



- (A) $(1, 2)$ (B) $(-1, 2)$ (C) $(-1, -2)$ (D) $(1, -2)$ (E) $(1, -1)$

Source: 2014 Gauss Grade 8 #15

Primary Topics: Geometry and Measurement

Secondary Topics: Coordinate Geometry | Transformations

Answer: B

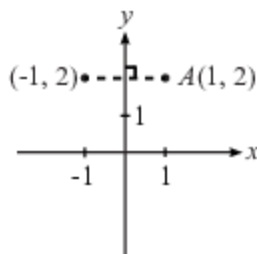
Solution:

After reflecting the point $A(1, 2)$ in the y -axis, the y -coordinate of the image will be the same as the y -coordinate of point A , $y = 2$.

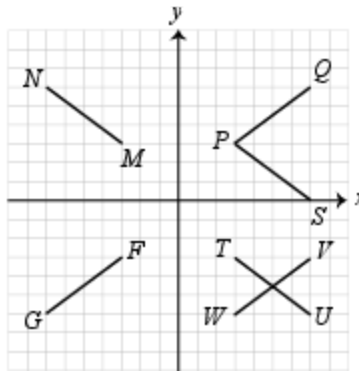
Point A is a distance of 1 to the right of the y -axis.

The image will be the same distance from the y -axis, but to the left of the y -axis. Thus, the image has x -coordinate -1 .

The coordinates of the reflected point are $(-1, 2)$.



9. In the graph shown, which of the following represents the image of the line segment PQ after a reflection across the x -axis?



- (A) PS (B) TU (C) MN (D) WV (E) FG

Source: 2015 Gauss Grade 8 #13

Primary Topics: Geometry and Measurement

Secondary Topics: Transformations

Answer: B

Solution:

To find the image of PQ , we reflect points P and Q across the x -axis, then join them.

Since P is 3 units above the x -axis, then the reflection of P across the x -axis is 3 units below the x -axis at the same x -coordinate.

That is, point T is the image of P after it is reflected across the x -axis.

Similarly, after a reflection across the x -axis, the image of point Q will be 6 units below the x -axis but have the same x -coordinate as Q .

That is, point U is the image of Q after it is reflected across the x -axis. Therefore, the line segment TU is the image of PQ after it is reflected across the x -axis.

10. A rectangle has side lengths 3 and 4. One of its vertices is at the point $(1, 2)$. Which of the following *could not* be the coordinates of one of its other vertices?
- (A) $(-3, -1)$ (B) $(1, -5)$ (C) $(5, -1)$ (D) $(-2, 6)$ (E) $(1, -1)$

Source: 2015 Gauss Grade 8 #20

Primary Topics: Geometry and Measurement

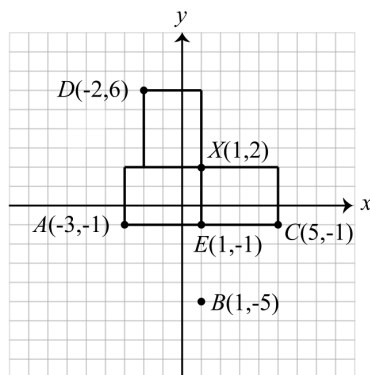
Secondary Topics: Coordinate Geometry

Answer: B

Solution:

Solution 1

Call the given vertex of the rectangle $(1, 2)$ point X and name each of the 5 answers to match the letters A through E , as shown.



Point $E(1, -1)$ is 3 units below point X (since their x -coordinates are equal and their y -coordinates differ by 3). Thus, $E(1, -1)$ could be the coordinates of a vertex of a 3 by 4 rectangle having vertex X (X and E would be adjacent vertices of the rectangle).

Point $C(5, -1)$ is 3 units below and 4 units right of point X (since their y -coordinates differ by 3 and their x -coordinates differ by 4). Thus, $C(5, -1)$ could be the coordinates of a vertex of a 3 by 4 rectangle having vertex X (X and C would be opposite vertices of the rectangle).

Point $A(-3, -1)$ is 3 units below and 4 units left of point X (since their y -coordinates differ by 3 and their x -coordinates differ by 4). Thus, $A(-3, -1)$ could be the coordinates of a vertex of a 3 by 4 rectangle having vertex X (X and A would be opposite vertices of the rectangle).

Point $D(-2, 6)$ is 4 units above and 3 units left of point X (since their y -coordinates differ by 4 and their x -coordinates differ by 3). Thus, $D(-2, 6)$ could be the coordinates of a vertex of a 3 by 4 rectangle having vertex X (X and D would be opposite vertices of the rectangle).

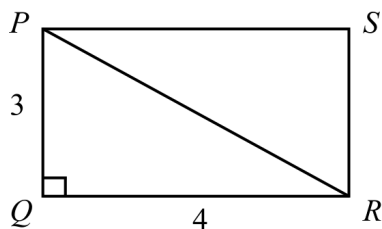
The only point remaining is $B(1, -5)$ and since it is possible for each of the other 4 answers to be one of the other vertices of the rectangle, then it must be $(1, -5)$ that can not be.

Point $B(1, -5)$ is 7 units below point X (since their y -coordinates differ by 7).

How might we show that no two vertices of a 3 by 4 rectangle are 7 units apart? (See Solution 2).

Solution 2

The distance between any two adjacent vertices of a 3 by 4 rectangle $PQRS$ is either 3 or 4 units (such as P and Q or Q and R , as shown).



The distance between any two opposite vertices of a rectangle (such as P and R) can be found using the Pythagorean Theorem.

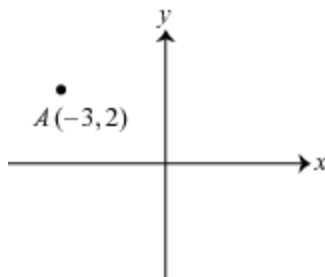
In the right-angled triangle PQR , we get $PR^2 = 3^2 + 4^2$ or $PR^2 = 9 + 16 = 25$ and so $PR = \sqrt{25} = 5$ (since $PR > 0$).

That is, the greatest distance between any two vertices of a 3 by 4 rectangle is 5 units.

As shown and explained in Solution 1, the distance between $X(1, 2)$ and $B(1, -5)$ is 7 units.

Therefore $(1, -5)$ could not be the coordinates of one of the other vertices of a 3 by 4 rectangle having vertex $X(1, 2)$.

11. A translation moves point $A(-3, 2)$ to the right 5 units and up 3 units. This translation is done a total of 6 times. After these translations, the point is at (x, y) . What is the value of $x + y$?



- (A) 34 (B) 49 (C) 53 (D) 47 (E) 43

Source: 2018 Gauss Grade 8 #17

Primary Topics: Geometry and Measurement

Secondary Topics: Transformations | Coordinate Geometry

Answer: D

Solution:

Solution 1:

Each translation to the right 5 units increases the x -coordinate of point A by 5. Similarly, each translation up 3 units increases the y -coordinate of point A by 3. After 1 translation, the point $A(-3, 2)$ would be at $B(-3 + 5, 2 + 3)$ or $B(2, 5)$. After 2 translations, the point $A(-3, 2)$ would be at $C(2 + 5, 5 + 3)$ or $C(7, 8)$. After 3 translations, the point $A(-3, 2)$ would be at $D(7 + 5, 8 + 3)$ or $D(12, 11)$. After 4 translations, the point $A(-3, 2)$ would be at $E(12 + 5, 11 + 3)$ or $E(17, 14)$. After 5 translations, the point $A(-3, 2)$ would be at $F(17 + 5, 14 + 3)$ or $F(22, 17)$. After 6 translations, the point $A(-3, 2)$ would be at $G(22 + 5, 17 + 3)$ or $G(27, 20)$. After these 6 translations, the point is at $(27, 20)$ and so $x + y = 27 + 20 = 47$.

Solution 2:

Each translation to the right 5 units increases the x -coordinate of point A by 5. Similarly, each translation up 3 units increases the y -coordinate of point A by 3. Therefore, each translation of point $A(-3, 2)$ to the right 5 units and up 3 units increases the sum of the x - and y -coordinates, $x + y$, by $5 + 3 = 8$. After 6 of these translations, the sum $x + y$ will increase by $6 \times 8 = 48$. The sum of the x - and y -coordinates of point $A(-3, 2)$ is $-3 + 2 = -1$. After these 6 translations, the value of $x + y$ is $-1 + 48 = 47$.
